4×4 Mesh NoC – BSV Results

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1. Microarchitecture

Each node has a five-port router (N, E, S, W, local PE). Packets are 268 bits, move as one unit (no flits) and use X–Y routing. Every port has a 2-entry input FIFO and a 2-entry output FIFO. One rule, `do_crossbar`, runs each cycle: it reads the head of every input FIFO, computes the output port from the destination, lets one arbiter per output pick a winner (only if that output FIFO has room), and moves the winners across. Link rules in `mkNoc` pass each output FIFO head to the neighbour's input FIFO, so a hop costs two cycles. Back-pressure is FIFO occupancy; nothing is dropped. One path per source–destination pair keeps packets of a pair in order, and forbidding Y-to-X turns makes the mesh deadlock-free.

2. Configurations

FIFO sizes and routing are the same in all three; only the output arbiter differs. Buffer storage is 20 entries × 268 bits = 5 360 bits per router in each.

Design	Arbiter	What changed	Flops
1	Round-robin	One 3-bit pointer per output; the search starts at the pointer, the first requester wins, the pointer moves past the winner. A waiting input is served within four grants.	15
2	Fixed, transit first	Fixed per-output order that puts traffic already in the network ahead of local injection (straight before local on E/W outputs; turning before straight on N/S outputs; local last). **Best design.**	0
3	Fixed, local first	One order for every output: local > N > E > S > W.	0

3. Table 2 for each design

L_0 and B_S come from directed tests on an idle network; B_N is 16 PEs sending 8-packet messages to random nodes (xorshift32, seed 42) over a 10 000-cycle window after a 200-cycle warm-up. Time units use the 1.0 ns clock at which all three designs were synthesised. The three designs share L_0, B_R and B_S, since nothing contends in those tests.

Design 1 – round-robin			buffer storage 5 360 bits/router	
Quantity	Estimate	Measured	In time units (1.0 ns)	Note
L_0	14 cycles	15 cycles	15 ns	
B_R	160 B/cycle	160 B/cycle*	160 GB/s	
B_S	32 B/cycle	17.07 B/cycle	17.07 GB/s	256 B in 15 cycles
B_N	≈170 B/cycle	183.79 B/cycle	183.79 GB/s	drained, PASS
Design 2 – fixed, transit first (best, evaluated)			buffer storage 5 360 bits/router	
L_0	14 cycles	15 cycles	15 ns	
B_R	160 B/cycle	160 B/cycle*	160 GB/s	
B_S	32 B/cycle	17.07 B/cycle	17.07 GB/s	256 B in 15 cycles
B_N	≈170 B/cycle	**191.21 B/cycle**	191.21 GB/s	drained, PASS
Design 3 – fixed, local first			buffer storage 5 360 bits/router	
L_0	14 cycles	15 cycles	15 ns	
B_R	160 B/cycle	160 B/cycle*	160 GB/s	
B_S	32 B/cycle	17.07 B/cycle	17.07 GB/s	256 B in 15 cycles
B_N	≈170 B/cycle	187.50 B/cycle	187.50 GB/s	drained, PASS

* B_R is the structural peak of five outputs × 32 B per cycle; it was not measured separately in simulation.

Estimates against measurements. L_0: seven routers at two cycles each give 14; the measured 15 is one cycle higher, most likely from the testbench's own input FIFO (not isolated). B_S: a link carrying one packet per cycle would give 32 B/cycle; the measurement is about half, one packet every two cycles. It is identical for all designs and nothing contends in the test, so the arbiter is not the cause; we did not isolate the cause. B_N: mean distance is 2.67 hops and a path of h hops takes about $2h + 3$ cycles, so a message of eight packets needs about 24 cycles, giving ≈170 B/cycle for sixteen PEs; the measured values are 8–12% higher, so the estimate was conservative. The network is far from saturated, so the arbiters differ by only about 4%.

4. Best design

Design 2 is submitted for evaluation. It has the highest B_N (+4.0% over Design 1, +2.0% over Design 3), the smallest area after synthesis (see `pd_results.pdf`) and no arbiter state. Its limitation, shared with Design 3, is that fixed priorities give no bound on waiting (G3); no run failed to drain. Design 1 is the only one with a bounded wait.